Geo Distributed Test Results

```
Outputs:

experiment_command = "python3 ../experiment.py --nodes http://18.233.244.123:8000 http://3.151.253.21:8000 http://52.41.20.26:8000 --rounds 100 --csv geo_distribut
ed_results.csv"
node_public_ips = {
  "nodeA" = "18.233.244.123"
  "nodeB" = "3.151.253.21"
  "nodeC" = "52.41.20.26"
}
node_regions = {
  "nodeA" = "us-east-1"
  "nodeB" = "us-east-2"
  "nodeC" = "us-west-2"
}
node_urls = [
  "http://18.233.244.123:8000",
  "http://3.151.253.21:8000",
  "http://52.41.20.26:8000",
]
private_key_path = "./raft-kv-geo-distributed.pem"
reset_command = "../reset_nodes.sh ./raft-kv-geo-distributed.pem 18.233.244.123 3.151.253.21 52.41.20.26"
status_urls = {
  "nodeA" = "http://18.233.244.123:8000/status"
  "nodeB" = "http://3.151.253.21:8000/status"
  "nodeC" = "http://52.41.20.26:8000/status"
}
```

```
Shreyans@LAPTOP-SN1NHDDA MINGW64 /d/College/CS6650 (BSDS Project)/Distributed-Raft-KV-Store/infrastructure/experiments/geo_distributed/terraform (main)
$ python ../experiment.py --nodes http://18.233.244.123:8000 http://3.151.253.21:8000 http://52.41.20.26:8000 --rounds 100 --csv geo_distributed_results.csv
==================================================================
Geo-distribution tradeoff experiment
==================================================================
nodes: http://18.233.244.123:8000, http://3.151.253.21:8000, http://52.41.20.26:8000
rounds: 100

[round 1] leader=http://18.233.244.123:8000 write=200 201.46ms read_avg=229.14976666652365ms stale=0/3 convergence=1066.4582000026712ms
[round 2] leader=http://18.233.244.123:8000 write=200 386.33ms read_avg=365.6754999998763ms stale=0/3 convergence=844.0850999977556ms
[round 3] leader=http://18.233.244.123:8000 write=200 83.01ms read_avg=495.48820000200067ms stale=0/3 convergence=1363.3754999973462ms
[round 4] leader=http://18.233.244.123:8000 write=200 481.31ms read_avg=346.7864999984158ms stale=0/3 convergence=890.6093999976292ms
[round 5] leader=http://18.233.244.123:8000 write=200 433.81ms read_avg=457.27560000038164ms stale=0/3 convergence=746.650100001716ms
[round 6] leader=http://18.233.244.123:8000 write=200 416.54ms read_avg=375.11559999984456ms stale=0/3 convergence=1023.4088999932283ms
[round 7] leader=http://18.233.244.123:8000 write=200 290.27ms read_avg=490.74549999689526ms stale=0/3 convergence=1836.4776000016718ms
[round 8] leader=http://18.233.244.123:8000 write=200 148.12ms read_avg=249.72526666776199ms stale=0/3 convergence=328.8152000022819ms
[round 9] leader=http://18.233.244.123:8000 write=200 81.49ms read_avg=134.3734000013986ms stale=0/3 convergence=342.16449999803444ms
[round 10] leader=http://18.233.244.123:8000 write=200 88.06ms read_avg=130.28770000043247ms stale=0/3 convergence=859.3718000047375ms
[round 11] leader=http://18.233.244.123:8000 write=200 424.85ms read_avg=340.90186666677863ms stale=0/3 convergence=1229.7584000004357ms
[round 12] leader=http://18.233.244.123:8000 write=200 322.33ms read_avg=457.408133331531ms stale=0/3 convergence=1229.9042000013287ms
[round 13] leader=http://18.233.244.123:8000 write=200 291.04ms read_avg=580.3659999995338ms stale=0/3 convergence=1373.077000003832ms
[round 14] leader=http://18.233.244.123:8000 write=200 334.80ms read_avg=302.2019999989425ms stale=0/3 convergence=1417.432700000063ms
[round 15] leader=http://18.233.244.123:8000 write=200 143.35ms read_avg=453.64993333350867ms stale=0/3 convergence=1594.8013000015635ms
[round 16] leader=http://18.233.244.123:8000 write=200 150.71ms read_avg=537.855866663449ms stale=0/3 convergence=928.097500000149ms
[round 17] leader=http://18.233.244.123:8000 write=200 110.80ms read_avg=293.3578666658529ms stale=0/3 convergence=608.0676999990828ms
[round 18] leader=http://18.233.244.123:8000 write=200 89.22ms read_avg=114.52723333301644ms stale=0/3 convergence=340.3470999983256ms
[round 19] leader=http://18.233.244.123:8000 write=200 97.81ms read_avg=108.18816666384616ms stale=0/3 convergence=369.98859999584965ms
[round 20] leader=http://18.233.244.123:8000 write=200 91.73ms read_avg=122.55300000106217ms stale=0/3 convergence=356.89759999513626ms
```

```
[round 80] leader=http://18.233.244.123:8000 write=200 116.79ms read_avg=218.90169999824138ms stale=0/3 convergence=989.8803000032785ms
[round 81] leader=http://18.233.244.123:8000 write=200 75.19ms read_avg=114.91733333241427ms stale=0/3 convergence=316.4923000039097ms
[round 82] leader=http://18.233.244.123:8000 write=200 103.93ms read_avg=112.58213333348976ms stale=0/3 convergence=361.02350000146544ms
[round 83] leader=http://18.233.244.123:8000 write=200 94.94ms read_avg=114.78509999869857ms stale=0/3 convergence=411.65490000275895ms
[round 84] leader=http://18.233.244.123:8000 write=200 84.54ms read_avg=113.19576666573994ms stale=0/3 convergence=330.2941000001738ms
[round 85] leader=http://18.233.244.123:8000 write=200 79.58ms read_avg=107.68750000230891ms stale=0/3 convergence=700.3417000014451ms
[round 86] leader=http://18.233.244.123:8000 write=200 333.50ms read_avg=335.01006666726124ms stale=0/3 convergence=1189.442800001416ms
[round 87] leader=http://18.233.244.123:8000 write=200 102.35ms read_avg=398.40766666748095ms stale=0/3 convergence=1113.4958999973605ms
[round 88] leader=http://18.233.244.123:8000 write=200 306.92ms read_avg=351.7833000005339ms stale=0/3 convergence=1262.3333000010462ms
[round 89] leader=http://18.233.244.123:8000 write=200 103.78ms read_avg=383.4148999982669ms stale=0/3 convergence=1215.7399999996414ms
[round 90] leader=http://18.233.244.123:8000 write=200 457.83ms read_avg=405.8268666655446ms stale=0/3 convergence=1075.9757999985595ms
[round 91] leader=http://18.233.244.123:8000 write=200 74.78ms read_avg=212.05823333245158ms stale=0/3 convergence=1171.6483999989578ms
[round 92] leader=http://18.233.244.123:8000 write=200 248.88ms read_avg=443.26666666408227ms stale=0/3 convergence=1426.212200000009ms
[round 93] leader=http://18.233.244.123:8000 write=200 352.77ms read_avg=427.07246666395804ms stale=0/3 convergence=1176.1349999942468ms
[round 94] leader=http://18.233.244.123:8000 write=200 783.47ms read_avg=455.2912333359321ms stale=0/3 convergence=1715.734400000656ms
[round 95] leader=http://18.233.244.123:8000 write=200 632.25ms read_avg=342.055466666352ms stale=0/3 convergence=1267.1917999978177ms
[round 96] leader=http://18.233.244.123:8000 write=200 686.52ms read_avg=341.0561333342533ms stale=0/3 convergence=1385.6228000004194ms
[round 97] leader=http://18.233.244.123:8000 write=200 568.83ms read_avg=337.0490999999068ms stale=0/3 convergence=1433.5970999964047ms
[round 98] leader=http://18.233.244.123:8000 write=200 304.17ms read_avg=500.5518666633482ms stale=0/3 convergence=736.6327000054298ms
[round 99] leader=http://18.233.244.123:8000 write=200 311.63ms read_avg=340.41916666804656ms stale=0/3 convergence=1721.8939000013052ms
[round 100] leader=http://18.233.244.123:8000 write=200 407.62ms read_avg=443.1814999989001ms stale=0/3 convergence=1186.136299998907ms

==================================================================
Summary
==================================================================
write latency: avg=311.34ms p50=305.43ms min=74.53ms max=966.20ms
read latency: avg=363.05ms p50=372.51ms min=107.69ms max=765.16ms
convergence: avg=1090.10ms p50=1154.44ms min=316.49ms max=1974.03ms
immediate stale reads: 0/300
csv: geo_distributed_results.csv
```

Geo Co-Located Test Results

```
Outputs:

experiment_command = "python3 ../experiment.py --nodes http://100.27.191.136:8000 http://3.93.182.242:8000 http://54.160.191.62:8000 --rounds 100 --csv colocated_r
esults.csv"
node_public_ips = {
  "nodeA" = "100.27.191.136"
  "nodeB" = "3.93.182.242"
  "nodeC" = "54.160.191.62"
}
node_urls = [
  "http://100.27.191.136:8000",
  "http://3.93.182.242:8000",
  "http://54.160.191.62:8000",
]
private_key_path = "./raft-kv-geo-colocated.pem"
reset_command = "../reset_nodes.sh ./raft-kv-geo-colocated.pem 100.27.191.136 3.93.182.242 54.160.191.62"
status_urls = {
  "nodeA" = "http://100.27.191.136:8000/status"
  "nodeB" = "http://3.93.182.242:8000/status"
  "nodeC" = "http://54.160.191.62:8000/status"
}
```

```
Shreyans@LAPTOP-SN1NHDDA MINGW64 /d/College/CS6650 (BSDS Project)/Distributed-Raft-KV-Store/infrastructure/experiments/geo_colocated/terraform (main)
$ python ../experiment.py --nodes http://100.27.191.136:8000 http://3.93.182.242:8000 http://54.160.191.62:8000 --rounds 100 --csv colocated_results.csv
==================================================================
Geo-distribution tradeoff experiment
==================================================================
nodes: http://100.27.191.136:8000, http://3.93.182.242:8000, http://54.160.191.62:8000
rounds: 100

[round 1] leader=http://100.27.191.136:8000 write=200 408.44ms read_avg=252.36989999878764ms stale=0/3 convergence=786.7786999995587ms
[round 2] leader=http://100.27.191.136:8000 write=200 329.19ms read_avg=250.26133333449252ms stale=0/3 convergence=1055.6529999958002ms
[round 3] leader=http://100.27.191.136:8000 write=200 358.95ms read_avg=444.03590000001714ms stale=0/3 convergence=1681.0056000031182ms
[round 4] leader=http://100.27.191.136:8000 write=200 291.27ms read_avg=496.69790000189096ms stale=0/3 convergence=972.6964000001317ms
[round 5] leader=http://100.27.191.136:8000 write=200 58.72ms read_avg=181.87369999941438ms stale=0/3 convergence=1338.4581999998773ms
[round 6] leader=http://100.27.191.136:8000 write=200 69.19ms read_avg=411.6368666655035ms stale=0/3 convergence=769.5715000008931ms
[round 7] leader=http://100.27.191.136:8000 write=200 275.48ms read_avg=414.67130000091856ms stale=0/3 convergence=704.0927999987616ms
[round 8] leader=http://100.27.191.136:8000 write=200 131.14ms read_avg=454.8053333322362ms stale=0/3 convergence=1224.6304000000237ms
[round 9] leader=http://100.27.191.136:8000 write=200 143.79ms read_avg=600.860233331332ms stale=0/3 convergence=1201.5895000004093ms
[round 10] leader=http://100.27.191.136:8000 write=200 394.52ms read_avg=315.230733336648ms stale=0/3 convergence=998.7709999986691ms
[round 11] leader=http://100.27.191.136:8000 write=200 111.96ms read_avg=158.50540000246838ms stale=0/3 convergence=1578.5321000003023ms
[round 12] leader=http://100.27.191.136:8000 write=200 983.19ms read_avg=459.5555999985663ms stale=0/3 convergence=668.9418000023579ms
[round 13] leader=http://100.27.191.136:8000 write=200 196.53ms read_avg=287.08496666513383ms stale=0/3 convergence=890.9749000013107ms
[round 14] leader=http://100.27.191.136:8000 write=200 339.73ms read_avg=420.89056666736724ms stale=0/3 convergence=778.4018000020296ms
[round 15] leader=http://100.27.191.136:8000 write=200 621.72ms read_avg=333.7783000000248ms stale=0/3 convergence=1036.7937999981223ms
[round 16] leader=http://100.27.191.136:8000 write=200 579.70ms read_avg=78.3155000002201ms stale=0/3 convergence=783.235999995668ms
[round 17] leader=http://100.27.191.136:8000 write=200 89.04ms read_avg=73.74596666583481ms stale=0/3 convergence=191.0617999965325ms
[round 18] leader=http://100.27.191.136:8000 write=200 68.19ms read_avg=66.14789999972952ms stale=0/3 convergence=178.18090000218945ms
[round 19] leader=http://100.27.191.136:8000 write=200 81.33ms read_avg=70.78279999889976ms stale=0/3 convergence=217.62410000519594ms
[round 20] leader=http://100.27.191.136:8000 write=200 66.49ms read_avg=68.0837999995371ms stale=0/3 convergence=700.2311000032932ms
```

```
[round 80] leader=http://100.27.191.136:8000 write=200 63.90ms read_avg=70.39540000065851ms stale=0/3 convergence=222.2939999992377ms
[round 81] leader=http://100.27.191.136:8000 write=200 149.93ms read_avg=345.9706333305803ms stale=0/3 convergence=1014.4736000002013ms
[round 82] leader=http://100.27.191.136:8000 write=200 348.35ms read_avg=397.6345666645405ms stale=0/3 convergence=906.1008999997284ms
[round 83] leader=http://100.27.191.136:8000 write=200 157.06ms read_avg=559.5288999999563ms stale=0/3 convergence=771.7296999981045ms
[round 84] leader=http://100.27.191.136:8000 write=200 684.26ms read_avg=209.7734000029353ms stale=0/3 convergence=1312.7603000029922ms
[round 85] leader=http://100.27.191.136:8000 write=200 183.51ms read_avg=327.55243333425216ms stale=0/3 convergence=1160.5058999994071ms
[round 86] leader=http://100.27.191.136:8000 write=200 67.89ms read_avg=321.93673333191936ms stale=0/3 convergence=661.1455999955069ms
[round 87] leader=http://100.27.191.136:8000 write=200 66.03ms read_avg=60.27516666411733ms stale=0/3 convergence=207.53130000230158ms
[round 88] leader=http://100.27.191.136:8000 write=200 92.16ms read_avg=65.01679999928456ms stale=0/3 convergence=254.7950000007404ms
[round 89] leader=http://100.27.191.136:8000 write=200 78.09ms read_avg=79.49299999745563ms stale=0/3 convergence=195.54369999968912ms
[round 90] leader=http://100.27.191.136:8000 write=200 68.10ms read_avg=72.41200000134995ms stale=0/3 convergence=219.27529999811668ms
[round 91] leader=http://100.27.191.136:8000 write=200 64.31ms read_avg=66.80770000093617ms stale=0/3 convergence=228.90989999723388ms
[round 92] leader=http://100.27.191.136:8000 write=200 73.95ms read_avg=74.72006666406135ms stale=0/3 convergence=226.6416000056779ms
[round 93] leader=http://100.27.191.136:8000 write=200 70.53ms read_avg=70.35469999876416ms stale=0/3 convergence=518.3602999968571ms
[round 94] leader=http://100.27.191.136:8000 write=200 482.14ms read_avg=307.3354000031638ms stale=0/3 convergence=1546.5165999994497ms
[round 95] leader=http://100.27.191.136:8000 write=200 302.21ms read_avg=466.22143333418836ms stale=0/3 convergence=1175.205300001835ms
[round 96] leader=http://100.27.191.136:8000 write=200 395.96ms read_avg=203.36883333463143ms stale=0/3 convergence=819.4584000011673ms
[round 97] leader=http://100.27.191.136:8000 write=200 317.44ms read_avg=610.8198666624958ms stale=0/3 convergence=997.0130000001518ms
[round 98] leader=http://100.27.191.136:8000 write=200 74.07ms read_avg=312.26760000330006ms stale=0/3 convergence=791.2563999998383ms
[round 99] leader=http://100.27.191.136:8000 write=200 129.52ms read_avg=306.90799999865703ms stale=0/3 convergence=1214.1971999953967ms
[round 100] leader=http://100.27.191.136:8000 write=200 695.13ms read_avg=433.97890000293654ms stale=0/3 convergence=879.4024000017089ms

==================================================================
Summary
==================================================================
write latency: avg=266.26ms p50=196.66ms min=55.05ms max=983.19ms
read latency: avg=251.60ms p50=247.21ms min=58.05ms max=614.48ms
convergence: avg=799.20ms p50=784.16ms min=178.18ms max=1936.42ms
immediate stale reads: 0/300
csv: colocated_results.csv
```

Comparison

```
Shreyans@LAPTOP-SN1NHDDA MINGW64 /d/College/CS6650 (BSDS Project)/Distributed-Raft-KV-Store/infrastructure/experiments (main)
$ python compare_geo_results.py \
  --colocated geo_colocated/terraform/colocated_results.csv \
  --distributed geo_distributed/terraform/geo_distributed_results.csv \
  --output geo_tradeoff_report.md
================================================================================
Geo-Distribution Tradeoff Results
================================================================================

Deployment           Rounds  Avg Write   P95 Write   Avg Read    Avg Conv.   Stale
--------------------------------------------------------------------------------
Co-located              100   266.26 ms   684.80 ms   251.60 ms   799.20 ms     0.00%
Geo-distributed         100   311.34 ms   686.90 ms   363.05 ms  1090.10 ms     0.00%

Ratios: geo-distributed / co-located
- Avg write latency: 1.17x
- Avg read latency: 1.44x
- Avg convergence: 1.36x

Wrote report: geo_tradeoff_report.md
```

Interpretation

The geo-distributed deployment had higher average write latency. This is expected because each write must be replicated through Raft and acknowledged by a majority. When replicas are spread across regions, the leader needs to communicate over longer network paths before a write can be committed.

Read latency was also higher in the geo-distributed run. In this experiment, reads were issued from the same client location to all replicas, so spreading replicas across regions did not provide a client-proximity benefit. Instead, the client had to contact farther-away replicas, increasing average read latency.

Convergence time was higher for the geo-distributed deployment. This suggests that even after the leader accepts a write, followers in other regions take longer to expose the latest value. This matches the expected tradeoff: geographic placement can improve locality for nearby clients, but it makes cross-region coordination slower.

The stale-read count was zero in both experiments. This means that in the sampled rounds, all replicas returned the latest value immediately after the write completed. This may be because the current Raft implementation applies committed entries quickly enough relative to the measurement interval. It does not mean stale reads are impossible; it only means this run did not observe them.

Takeaways

The experiment supports the expected geo-distribution tradeoff:

- Co-located replicas provide lower write latency and faster convergence.
- Geo-distributed replicas increase coordination cost.
- Geo-distribution did not improve read latency in this setup because the client was not placed near each remote region.
- To demonstrate the read-locality benefit more clearly, future runs should place clients in multiple regions and compare each client's nearest follower against the leader region.

Limitations

This experiment is useful as a first comparison, but it has some limitations:

- The client was not deployed in multiple regions, so the read-locality benefit of geo-distribution was not fully tested.
- The script measures immediate follower visibility from one client machine, not from clients near each follower.
- The current system uses a simplified Raft implementation and does not yet include production-grade log persistence or optimized follower-read semantics.
- The experiment used 100 sequential rounds, not a high-concurrency Locust workload.

Future versions can improve this by running regional clients, separating leader reads from follower reads, and measuring throughput under concurrent load.